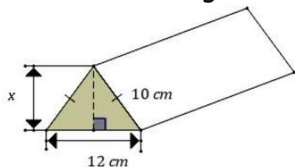


(A) Showcasing CoT and DRT Reasoning Paradigms

Find total surface area of the triangular prism shown. The height is 16 cm.



Qwen3-VL CoT

Prediction: **960** ✗ | 4,937 tokens

▀ verbose filler ▀ guessed premise

Got it, let's try to figure out the total surface area... First, I need to recall what a triangular prism is... First, let's list out the given information... Let me check the diagram... Let's clarify....

[omitted: many repeated trials and reinterpretations] Maybe the height of the triangular base is 16 cm... area of the base is 96, two bases = 192. Lateral surface area = $48 \times 16 = 768$. Total = $192 + 768 = 960$. [omitted: repeated double-checks that still miss the true answer]

Dense Reasoning Trace

Prediction: **608** ✓ | 137 tokens

Visual

Triangular prism; base = 12 cm; equal sides = 10 cm; prism length = 16 cm.

Think

[Priors] isosceles altitude bisects base;
Pythagorean theorem: $a^2 + b^2 = c^2$
→ half-base = 6
→ $x = \sqrt{10^2 - 6^2} = 8$
→ triangle area = $\frac{1}{2} \times 12 \times 8 = 48$
→ two bases = 96
→ lateral area = $32 \times 16 = 512$
→ total surface area = 608.

Answer
608

(B) Accuracy-Token Tradeoff Across Reasoning Methods

